

Stochastic Transitive Reinforcement Learning: Calibrated Divide-and-Conquer Value Propagation under Stochastic Dynamics

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Abstract

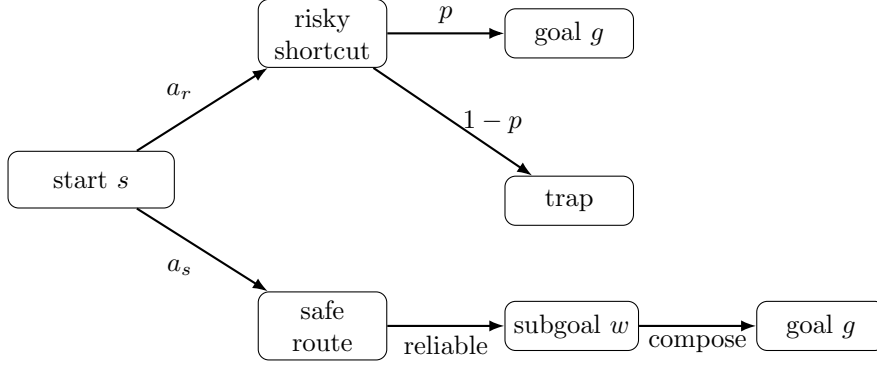
Long-horizon offline goal-conditioned reinforcement learning requires propagating sparse reachability information across many decision steps. Transitive RL (TRL) shows that deterministic goal-reaching problems contain a divide-and-conquer structure: if an agent can reach an intermediate subgoal and then reach the final goal, long paths can be propagated in logarithmically many transitive sweeps. This deterministic composition is not calibrated in stochastic environments. A lucky rollout through a risky shortcut or teleporter is only one outcome from a transition distribution, but deterministic realized-path composition can treat it as a reliable edge. We introduce *Stochastic Transitive RL*, a calibrated transitive value-iteration algorithm that composes stochastic reachability probabilities instead of binary realized path labels. Each update combines an empirical Bellman expectation backup, which calibrates stochastic outcomes, with a transitive product backup, which propagates reliable long-horizon reachability. In tabular stochastic shortcut tasks and 2D stochastic grid shortcuts, Stochastic TRL learns safe long-horizon policies with logarithmic matched sweep budgets, where matched Bellman and realized-positive baselines choose risky shortcuts. On OGBench PointMaze teleport navigate and stitch diagnostics, Stochastic TRL reaches 0.901 mean success with a 6-sweep budget, matching a 180-sweep Bellman reference while matched Bellman reaches 0.343. On OGBench AntMaze teleport with learned BC executors, Stochastic TRL reaches 0.947 success on navigate and 0.960 on stitch, again matching the long-sweep Bellman reference while matched Bellman remains near 0.31. Across three independently trained AntMaze controllers for each task, Stochastic TRL remains above 0.933 success on navigate and above 0.950 success on stitch. In a learned high-level module screen that replaces both table transition fitting and table action values with raw-observation MLP heads, Stochastic TRL still reaches 0.933 success on AntMaze navigate and 0.967 on AntMaze stitch across three evaluation seeds. A controlled learned-critic screen further shows that a small MLP trained with the stochastic transitive target solves all safe-optimal stochastic shortcut settings, while matched neural Bellman and support-only transitive critics fail.

1 Introduction

Goal-conditioned reinforcement learning (GCRL) asks an agent to reach arbitrary goal states. In the offline setting, this problem is particularly sensitive to long-horizon value propagation: a Bellman backup transmits sparse reachability information one transition at a time, so a path of length H requires roughly H value-iteration sweeps before the start state sees the goal. Transitive RL [4] addresses this in deterministic environments by turning a triangle-inequality structure into a divide-and-conquer value update. If a policy can reach w from s and then reach g from w , a transitive backup can compose these two shorter subproblems and propagate length- H structure in $O(\log H)$ sweeps.

The same idea is not directly valid in stochastic environments. A successful trajectory segment through a stochastic teleporter or risky shortcut is not a reliable transition; it is one sampled outcome from a transition distribution. Composing such samples as if they were deterministic edges creates optimistic plans that exploit lucky outcomes. This failure mode is directly relevant to modern offline GCRL benchmarks such as OGBench, whose tasks include long-horizon stitching and stochastic teleport dynamics [3].

This paper studies that tension and proposes Stochastic Transitive RL. The central change is simple: compose calibrated stochastic reachability estimates, not realized path labels. The resulting operator uses a Bellman expectation backup to estimate the probability of reaching subgoals under stochastic dynamics, and



Bellman calibration values the shortcut near $\gamma^d p$;
transitive composition propagates the reliable safe route in $O(\log H)$ sweeps.

Figure 1: Stochastic Transitive RL composes calibrated reachability probabilities. A successful risky trajectory is not treated as a deterministic edge; reliable route segments are still composed divide-and-conquer style.

a transitive product backup to propagate long-horizon routes. The Bellman term keeps stochastic shortcuts calibrated; the transitive term keeps TRL’s horizon efficiency on reliable paths.

Our contributions are:

1. We identify a stochastic failure mode of deterministic or realized-path TRL: lucky stochastic outcomes can be over-composed into high-value shortcuts.
2. We introduce a calibrated stochastic transitive backup for discounted goal-reaching values.
3. We prove lower-bound preservation for the empirical-model operator and logarithmic propagation on reliable paths.
4. We show controlled tabular and grid evidence that Stochastic TRL preserves TRL-style horizon efficiency while avoiding risky shortcut optimism.
5. We provide OGBench PointMaze and AntMaze teleport diagnostics showing high long-horizon success under a matched 6-sweep budget, including three-controller AntMaze robustness on both navigate and stitch.
6. We give learned high-level module diagnostics showing that raw-observation transition and tie-preserving policy heads preserve the AntMaze hard-task gain over matched Bellman.
7. We include a controlled learned-critic diagnostic in which an MLP stochastic-TRL critic solves stochastic long-horizon shortcut tasks while neural Bellman and support-only critics fail under the same budget.

2 Problem Setting

Let $\mathcal{S}$ and $\mathcal{A}$ be finite state and action sets. For a fixed goal g , define the discounted hitting value after taking first action a in state s :

$$Q^*(s, a, g) = \sup_{\pi} \mathbb{E} [\gamma^{\tau_g} \mathbf{1}\{\tau_g < \infty\} \mid s_0 = s, a_0 = a], \quad (1)$$

where $\tau_g = \inf\{t \geq 0 : s_t = g\}$ and $\gamma \in (0, 1)$. We use the boundary convention $Q^*(g, a, g) = 1$ and $V^*(s, g) = \max_a Q^*(s, a, g)$.

Our prototype experiments use empirical transition or topology models $\hat{P}(s'|s, a)$ for high-level planning. Continuous-control experiments then execute the high-level plan with a low-level controller: a simple topology executor for PointMaze and a learned goal-conditioned behavior-cloning (BC) executor for AntMaze. This

isolates the value-propagation question: can calibrated transitive backups learn useful long-horizon plans under stochastic dynamics faster than matched Bellman updates?

3 Method

For any candidate table Q , let $V_Q(s, g) = \max_a Q(s, a, g)$, with $V_Q(g, g) = 1$.

Bellman calibration. Given an empirical transition model $\hat{P}$, define

$$(\mathcal{B}Q)(s, a, g) = \gamma \mathbb{E}_{s' \sim \hat{P}(\cdot|s, a)} V_Q(s', g), \quad s \neq g, \quad (2)$$

$$(\mathcal{B}Q)(g, a, g) = 1. \quad (3)$$

This term calibrates stochastic outcomes. Entering a teleporter does not become a deterministic edge to every observed exit; it receives the expected reachability value under the empirical exit distribution.

Transitive composition. The transitive candidate composes calibrated reachability through an intermediate subgoal:

$$(\mathcal{T}Q)(s, a, g) = \max_{w \neq s} Q(s, a, w) V_Q(w, g), \quad s \neq g, \quad (4)$$

$$(\mathcal{T}Q)(g, a, g) = 1. \quad (5)$$

The exclusion $w \neq s$ prevents the tautology $Q(s, a, s) = 1$ from copying $V_Q(s, g)$ into every first action.

Stochastic TRL update. The algorithm starts from one-step empirical reachability,

$$Q_0(s, a, g) = \gamma \hat{P}(g|s, a), \quad Q_0(g, a, g) = 1, \quad (6)$$

and iterates the pointwise maximum

$$Q_{k+1} = \max(\mathcal{B}Q_k, \mathcal{T}Q_k). \quad (7)$$

The induced policy is $\pi_k(s, g) = \arg \max_a Q_k(s, a, g)$.

Algorithm 1: Stochastic Transitive Value Iteration. Input empirical transition model $\hat{P}$, discount γ , sweep budget K . Initialize $Q_0(s, a, g) = \gamma \hat{P}(g|s, a)$ and $Q_0(g, a, g) = 1$. For $k = 0, \dots, K - 1$, compute $V_k(s, g) = \max_a Q_k(s, a, g)$, then update each non-goal triple with Eq. (7). Return the greedy policy $\arg \max_a Q_K(s, a, g)$.

4 Theory

Theorem 1 (Lower-bound preservation). *Fix a transition model P and let Q^* be the optimal discounted hitting value for that model. If $0 \leq Q(s, a, g) \leq Q^*(s, a, g)$ for all s, a, g , then*

$$\mathcal{B}Q \leq Q^*, \quad \mathcal{T}Q \leq Q^*, \quad \max(\mathcal{B}Q, \mathcal{T}Q) \leq Q^*.$$

Consequently, if $Q_0 \leq Q^$, every iterate of Eq. (7) is a pointwise lower bound on Q^* .*

Proof. Bellman monotonicity gives $V_Q(s, g) \leq V^*(s, g)$ for all s, g , and hence

$$(\mathcal{B}Q)(s, a, g) \leq \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} V^*(s', g) = Q^*(s, a, g).$$

For the transitive term, fix an intermediate state $w \neq s$. The product $Q^*(s, a, w)V^*(w, g)$ is the value contribution of a feasible composite policy: after taking a , first try to hit w , then, if w is reached, switch to an optimal policy for g from w . If g is reached before w , the true hitting value for g can only be larger than this contribution. Since $Q^*(s, a, g)$ is optimal over all policies for goal g , $Q^*(s, a, w)V^*(w, g) \leq Q^*(s, a, g)$. Using $Q \leq Q^*$ and $V_Q \leq V^*$ gives $Q(s, a, w)V_Q(w, g) \leq Q^*(s, a, g)$. Maximizing over $w \neq s$ preserves the inequality. $\square$

Table 1: Planning-budget curve on the hardest 16x8 stochastic grid. Stochastic TRL reaches the safe long-horizon route with 7 sweeps; Bellman requires 126 sweeps. Source: `grid_budget_curve.csv`.

Method	Sweeps	Success	Safe action	Portal action
Bellman	8	0.048	0.000	1.000
Bellman	120	0.048	0.000	1.000
Bellman	126	1.000	1.000	0.000
Stochastic TRL	6	0.048	0.000	1.000
Stochastic TRL	7	1.000	1.000	0.000

Theorem 2 (Logarithmic propagation on reliable paths). *Consider a deterministic path $s_0 \rightarrow s_1 \rightarrow \dots \rightarrow s_H$ with goal $g = s_H$ and path actions a_i . If the initialization contains exact one-step values $Q_0(s_i, a_i, s_{i+1}) = \gamma$ and $Q_0(g, a, g) = 1$, then after k stochastic TRL sweeps every path segment of length at most 2^k has exact value $Q_k(s_i, a_i, s_j) = \gamma^{j-i}$. Thus a reliable path of length H can be represented after $\lceil \log_2 H \rceil$ transitive sweeps.*

Proof. The claim holds for $k = 0$ because the initialization contains every length-one edge. Assume it holds for all segments of length at most $m = 2^k$. For a segment from s_i to s_j of length at most $2m$, choose an intermediate state s_ℓ so both subsegments have length at most m . By induction, the first subsegment has value $\gamma^{\ell-i}$ and the second has value $\gamma^{j-\ell}$. The transitive backup composes them into γ^{j-i} . The lower-bound theorem prevents overshooting the deterministic hitting value, so the value is exact. $\square$

Proposition 1 (Shortcut calibration). *In a stochastic shortcut MDP, suppose a safe route reaches g deterministically after H steps, while a risky route reaches g after d steps with probability p and otherwise enters a trap. The calibrated risky value is $\gamma^d p$, while the safe value is γ^H . If $\gamma^H > \gamma^d p$, the calibrated optimal first action is safe. A realized-positive transitive estimator that conditions only on successful risky trajectories can estimate the risky branch near γ^d , overestimating by a factor of approximately $1/p$.*

5 Experiments

All reported claims in this section are generated from CSV artifacts and checked by the repository verifier. Unless otherwise stated, “matched” means the same 6-sweep planning budget for Bellman and Stochastic TRL, while “full” means a 180-sweep Bellman reference.

5.1 Controlled stochastic shortcut tasks

Safe path lengths range from 16 to 128. In safe-optimal settings, Stochastic TRL reaches 1.000 success with matched logarithmic sweeps, while matched Bellman and MC-positive baselines choose the risky shortcut and succeed only near the shortcut probability. On the hardest 16x8 stochastic grid, whose safe path length is 127, Stochastic TRL reaches 1.000 success after 7 sweeps; Bellman remains at portal-rate success through 120 sweeps and first reaches 1.000 success at 126 sweeps (Table 1).

We also train small neural critics on finite stochastic shortcut MDPs from offline empirical transitions. This controlled screen keeps the same safe-optimal shortcut regimes but replaces table values with MLP critics. Neural Stochastic TRL solves every safe-optimal setting and makes the correct safe first-action decision in all 12 conditions, while matched neural Bellman propagates too slowly and neural support TRL over-composes the lucky risky shortcut (Table 2).

5.2 OGBench PointMaze and AntMaze teleport

PointMaze uses a dataset-inferred topology scaffold and a simple topology executor. AntMaze uses the same topology-level stochastic planner with a learned goal-conditioned BC executor. Waypoint goals use full future observations and 16 body-nearest candidates so target waypoints are compatible with the current ant body state.

Table 2: Learned finite-MDP critic screen on stochastic risky-shortcut tasks. Rows average over 12 safe-optimal settings with safe lengths 16, 32, and 64. Source: `neural_shortcut_phase.csv`.

Method	Success	Correct decision	Risky action
Neural Bellman TD	0.027	0.583	0.417
Neural Stochastic TRL	1.000	1.000	0.000
Neural Support TRL	0.093	0.000	1.000
Table Stochastic TRL	1.000	1.000	0.000
Table full Bellman	1.000	1.000	0.000

Table 3: Main long-horizon teleport results. Stochastic TRL reaches the long-sweep Bellman reference at the matched 6-sweep budget. PointMaze rows use five evaluation seeds and 50 episodes per task; AntMaze rows use three evaluation seeds and 20 episodes per task. Source: `main_hard_task_results.csv`.

Environment	Executor	Bellman matched	Stochastic TRL	Full Bellman	Gain
PointMaze navigate	topology scaffold	0.343	0.901	0.901	0.558
PointMaze stitch	topology scaffold	0.343	0.901	0.901	0.558
AntMaze navigate	BC executor	0.310	0.947	0.947	0.637
AntMaze stitch	BC executor	0.317	0.960	0.960	0.643

Table 3 is the central empirical result. On all four teleport diagnostics, Stochastic TRL matches the 180-sweep Bellman reference using only 6 sweeps. Matched Bellman remains local, with mean success at 0.343 on PointMaze and about 0.31 on AntMaze.

5.3 Calibration ablation

To isolate stochastic calibration from transitive composition alone, we compare against a deterministic support-TRL baseline on PointMaze teleport stitch. This baseline treats every observed stochastic transition support edge as reliable. It improves over matched Bellman but remains far below calibrated Stochastic TRL (Table 4).

5.4 AntMaze controller robustness

Because AntMaze results combine a high-level planner with a learned executor, we evaluate robustness across independently trained task-specific BC controllers. Navigate uses 50k-step controllers; stitch uses 20k-step controllers. For both tasks, Stochastic TRL remains above 0.933 success across all controller seeds and stays close to the long-sweep Bellman reference (Table 5).

5.5 Learned high-level module diagnostic

The headline continuous-control tables isolate value propagation with a topology-level planner. To test whether the signal survives learned high-level modules, we replace table transition fitting with a raw-observation MLP jump-change transition head and replace table action values with a tie-preserving raw-observation policy head. The executor remains the saved full-goal BC controller, and planning is still over the high-level cell abstraction. On the hard AntMaze tasks, the learned-module screen preserves the stochastic TRL advantage across three evaluation seeds and matches the 180-sweep Bellman reference (Table 6). Appendix Table 16 also reports robustness across three transition seeds.

Table 4: PointMaze teleport stitch support ablation. Deterministic support TRL helps but remains far below calibrated Stochastic TRL, showing that stochastic calibration is essential. Source: `pointmaze_stitch_support_baseline_5seed.csv`.

Method	Sweeps	Mean success
Bellman matched	6	0.343
Support TRL	6	0.449
Stochastic TRL	6	0.901
Bellman full	180	0.901

Table 5: AntMaze controller-seed robustness. Each controller seed is evaluated with three evaluation seeds and 20 episodes per task. Source: generated navigate and stitch controller-seed tables.

Task	Controller seed	Bellman matched	Stochastic TRL	Bellman full
Navigate	0	0.310	0.947	0.947
Navigate	1	0.320	0.933	0.933
Navigate	2	0.310	0.940	0.940
Stitch	0	0.317	0.960	0.960
Stitch	1	0.323	0.950	0.950
Stitch	2	0.323	0.963	0.967

6 Related Work

Our work builds directly on Transitive RL [4], which introduced divide-and-conquer value learning for deterministic offline GCRL. Stochastic TRL keeps the transitive structure but replaces deterministic realized composition with an empirical stochastic Bellman calibration term. The benchmark setting follows OGBench [3], a benchmark designed to test long-horizon reasoning, stitching, high-dimensional inputs, and stochasticity in offline GCRL. More broadly, our experiments sit within offline RL and offline GCRL, where datasets such as D4RL [1] and value-learning methods such as IQL [2] have shown the importance of conservative value estimation under distribution shift. Our method is complementary to learned-controller approaches: in continuous control, we use learned BC executors to isolate the high-level stochastic value-propagation problem.

7 Limitations

The continuous-control results are learned-controller, high-level abstraction diagnostics, not yet a complete end-to-end neural implementation of Stochastic TRL. The strongest AntMaze module screen learns raw-observation transition and tie-preserving policy heads, but planning still occurs over a cell abstraction and execution is isolated through a learned BC controller. This design is deliberate for algorithmic diagnosis, but the present results should not be read as solving a full neural replacement for TRL. The theory is also stated for an exact empirical transition model. With finite data, the lower-bound statement is with respect to the empirical model rather than necessarily the true environment.

8 Conclusion

Stochastic TRL extends TRL’s divide-and-conquer horizon propagation to stochastic environments by composing calibrated reachability probabilities. The Bellman term prevents lucky stochastic outcomes from becoming reliable edges; the transitive term retains logarithmic propagation on reliable long-horizon routes. Across controlled stochastic shortcuts, PointMaze teleport, and AntMaze teleport diagnostics, the method reaches high success with a small matched planning budget and matches or nearly matches a long-sweep Bellman reference. Learned raw-observation high-level transition and policy heads preserve the AntMaze hard-

Table 6: AntMaze learned high-level module screen. A raw-observation MLP transition head and raw-observation tie-preserving policy head preserve the hard-task advantage across three evaluation seeds. Source: `results/paper_tables/antmaze_rawobs_transition_tie_policy_head_ep10_evalseed012.csv`.

Environment	Controller	Evaluation	Bellman matched	Stochastic TRL	Full Bellman	Transition top-1	Action agreement
AntMaze navigate	50k full-goal BC + body-nearest k16	3 eval seeds, tasks 4-5, 10 episodes/task	0.283	0.933	0.933	1.000	0.999
AntMaze stitch	20k full-goal BC + body-nearest k16	3 eval seeds, tasks 4-5, 10 episodes/task	0.283	0.967	0.967	1.000	0.999

task gain, suggesting a path toward neural stochastic TRL. The next step is to replace the remaining high-level cell abstraction with an end-to-end learned stochastic transitive critic while preserving the calibration and robustness demonstrated here.

A Verification Protocol

All headline claims are checked by `scripts/verify_main_claims.py`. The verifier requires:

- matched Bellman and Stochastic TRL use 6 sweeps;
- full Bellman uses 180 sweeps;
- Stochastic TRL success is at least 0.90 on every headline row;
- matched Bellman success is at most 0.40 on every headline row;
- Stochastic TRL improves over matched Bellman by at least 0.50 on every headline row;
- AntMaze controller-seed aggregates have 20 episodes per task, three evaluation seeds, correct BC-step metadata, and full-Bellman gaps at most 0.02;
- learned-transition screens match their raw source CSVs, use offline cell-change targets or 20 sampled outcomes per state-action row, and keep Stochastic TRL at least 0.90 success on PointMaze and 0.95 success on AntMaze hard tasks.

B Hard-Task Stress Checks

Table 7 shows short single-evaluation-seed stress checks used for fast iteration on harder task slices. These rows are not a replacement for the multi-seed headline table, but they support debugging and stress-test interpretation.

Table 7: Fast hard-task stress checks. Source: `results/paper_tables/hard_task_stress_seed0.csv`.

Environment	Task scope	Bellman matched	Support TRL	Stochastic TRL	Full Bellman
PointMaze stitch	tasks 4,5	0.420	0.480	0.930	0.930
AntMaze navigate	tasks 4,5	0.300	–	0.950	0.950
AntMaze stitch	tasks 4,5	0.350	–	1.000	1.000

C Focused PointMaze Single-Task Check

Table 8 reports a higher-episode focused check on PointMaze teleport stitch task 5. This row is appendix evidence for absolute success on a single long-horizon task, not a replacement for the all-task headline table.

Table 8: Focused PointMaze single-task check. Source: generated task-5 PointMaze table.

Environment	Task	Evaluation	Bellman matched	Stochastic TRL	Full Bellman
PointMaze stitch	5	5 eval seeds, 100 episodes	0.380	0.908	0.908

Table 9: Learned-transition screens. Sources: `results/paper_tables/pointmaze_learned_transition.csv` and `results/paper_tables/antmaze_learned_transition_robustness.csv`.

Environment	Model	Evaluation	Bellman matched	Stochastic TRL	Full Bellman
PointMaze navigate	table softmax	seed 0, 50 episodes/task	0.408	0.916	0.916
PointMaze stitch	table softmax	seed 0, 50 episodes/task	0.408	0.916	0.916
PointMaze navigate	raw obs MLP	seed 0, 3 transition seeds, 50 episodes/task	0.512	0.916	0.916
PointMaze stitch	raw obs MLP	seed 0, 3 transition seeds, 50 episodes/task	0.483	0.916	0.916
AntMaze navigate	learned transition + BC	3 transition seeds, 10 episodes/task	0.300	0.950	0.950
AntMaze stitch	learned transition + BC	3 transition seeds, 10 episodes/task	0.367	1.000	1.000
AntMaze navigate	raw obs MLP + BC	3 transition seeds, 10 episodes/task	0.300	0.950	0.950
AntMaze stitch	raw obs MLP + BC	3 transition seeds, 10 episodes/task	0.300	1.000	1.000

D Learned Transition Screens

Table 9 reports learned-transition screens. The PointMaze rows include both a table-softmax high-level transition model and a raw-observation MLP transition head trained from collapsed offline cell-change targets. The AntMaze rows include finite-sample table fitting with learned BC executors and raw-observation MLP jump-change transition heads. These rows support the learned stochastic modeling story but remain appendix screens rather than replacements for the multi-evaluation-seed headline table.

Table 10 is a stricter learned value/control diagnostic for PointMaze. It keeps the learned raw-observation transition head but replaces table action values with a raw-observation policy head that preserves ties among near-optimal high-level actions. This head recovers the successful sticky-action semantics used by the executor on both teleport variants; scalar value and previous-action-free single-label policy heads fail in the same screen. Table 12 shows that an explicit previous high-level action is enough to make a conventional single-label policy head recover the same behavior.

Table 11 tests the same PointMaze high-level MDP with a learned joint-action critic rather than only a policy head. The critic maps raw-observation state-goal features and state/goal identifiers to all four high-level action values. We first generate a matched-budget stochastic transitive target from the learned transition model, then fit the critic to that generated target. This learned critic recovers the exact-model policy on both teleport variants; a matched-budget Bellman TD critic with the same architecture remains low. With the same proportional PointMaze controller used in the topology planner, the generated-target q-head also reaches 0.933 all-task real-environment success across evaluation seeds 0, 1, and 2, and 1.00 success on hard tasks 4 and 5 in the seed-0 screen, for both teleport variants. Fully self-bootstrapped neural fitted iteration remains a limitation.

Table 13 repeats the same PointMaze raw-observation transition plus tie-policy screen over three evaluation seeds. The aggregate remains above 0.90 success on both teleport variants and matches the long-sweep Bellman reference.

Table 14 fixes the evaluation seed and repeats the combined raw-observation transition plus tie-policy screen over three transition-model seeds. This isolates transition-fit sensitivity from rollout randomness.

Table 15 repeats the tie-preserving high-level policy-head diagnostic on AntMaze hard tasks with the learned BC executor and topology transition model. This is a single-seed screen over tasks 4 and 5, not a replacement for the multi-seed AntMaze topology table.

Table 16 combines the learned high-level modules on AntMaze: a raw-observation MLP jump-change transition head and a raw-observation tie-preserving policy head. This screen still uses the cell abstraction and learned BC executor, but removes both table transition fitting and table action values, and repeats the transition fit over three transition seeds. We also report an evaluation-seed aggregate with transition seed 0

Table 10: PointMaze tie-preserving policy-head screen. Source: [results/paper_tables/pointmaze_tie_policy_head_ep20_seed0.csv](#).

Environment	Control head	Evaluation	Bellman matched	Stochastic TRL	Full Bellman	Action agreement
PointMaze navigate	raw-observation tie-policy MLP	seed 0, all tasks, 20 episodes/task	0.530	0.980	0.980	0.982
PointMaze stitch	raw-observation tie-policy MLP	seed 0, all tasks, 20 episodes/task	0.530	0.980	0.980	0.978

Table 11: PointMaze joint-action critic diagnostic. Exact model evaluation over all five tasks with a learned raw-observation transition head; the verifier also checks matching multi-seed and hard-slice real-environment rollout rows.

Environment	Q-head Bellman TD	Q-head Stochastic target	Table Bellman	Table Stochastic	Q-head action agreement
PointMaze navigate	0.066	1.000	0.476	1.000	0.966
PointMaze stitch	0.072	1.000	0.481	1.000	0.959

fixed.

The previous-action-conditioned policy head in Table 17 is a stricter single-label variant of the high-level control head. It preserves the successful table tie-breaking behavior by conditioning on the previous high-level action.

Table 18 replaces the PointMaze topology executor with saved full-goal BC controllers. The high-level value computation is unchanged, so this screen isolates whether the PointMaze gain depends on the simple topology executor.

Table 19 compares this waypoint execution path to short direct final-goal actor screens in the official neural training loop. The direct screens are useful negative controls: they show that low success in the official actor path can be a controller bottleneck rather than evidence against the high-level stochastic value update.

References

- [1] Justin Fu, Aviral Kumar, Ofir Nachum, George Tucker, and Sergey Levine. D4RL: Datasets for deep data-driven reinforcement learning. In *Advances in Neural Information Processing Systems*, 2020.
- [2] Ilya Kostrikov, Ashvin Nair, and Sergey Levine. Offline reinforcement learning with implicit Q-learning. In *International Conference on Learning Representations*, 2022.
- [3] Seohong Park, Kevin Frans, Benjamin Eysenbach, and Sergey Levine. OGBench: Benchmarking offline goal-conditioned RL, 2024.
- [4] Seohong Park, Aditya Oberai, Pranav Atreya, and Sergey Levine. Transitive RL: Value learning via divide and conquer. In *International Conference on Learning Representations*, 2026. arXiv:2510.22512.

Table 12: PointMaze previous-action policy-head screen. Source: [results/paper_tables/pointmaze_rawobs_transition_prev_policy_head_ep20_seed0.csv](#).

Environment	Control head	Evaluation	Bellman matched	Stochastic TRL	Full Bellman	Action agreement
PointMaze navigate	previous-action policy MLP	seed 0, all tasks, 20 episodes/task	0.530	0.980	0.980	1.000
PointMaze stitch	previous-action policy MLP	seed 0, all tasks, 20 episodes/task	0.530	0.980	0.980	1.000

Table 13: PointMaze tie-policy eval-seed robustness.

Environment	Control head	Evaluation	Bellman matched	Stochastic TRL	Full Bellman	Action agreement
PointMaze navigate	raw-observation tie-policy MLP	3 eval seeds, all tasks, 20 episodes/task	0.417	0.927	0.927	0.982
PointMaze stitch	raw-observation tie-policy MLP	3 eval seeds, all tasks, 20 episodes/task	0.417	0.927	0.927	0.978

Table 14: PointMaze tie-policy transition-seed robustness.

Environment	Control head	Evaluation	Bellman matched	Stochastic TRL	Full Bellman	Action agreement
PointMaze navigate	raw-observation tie-policy MLP	seed 0, 3 transition seeds, all tasks, 20 episodes/task	0.530	0.980	0.980	0.978
PointMaze stitch	raw-observation tie-policy MLP	seed 0, 3 transition seeds, all tasks, 20 episodes/task	0.530	0.980	0.980	0.978

Table 15: AntMaze tie-preserving policy-head hard-task screen. Source: [results/paper_tables/antmaze_tie_policy_head_hard_tasks_ep20_seed0.csv](#).

Environment	Controller	Evaluation	Bellman matched	Stochastic TRL	Full Bellman	Action agreement
AntMaze navigate	50k full-goal BC + body-nearest k16	seed 0, tasks 4-5, 20 episodes/task	0.350	0.925	0.925	1.000
AntMaze stitch	20k full-goal BC + body-nearest k16	seed 0, tasks 4-5, 20 episodes/task	0.400	0.950	0.950	1.000

Table 16: AntMaze raw-observation transition plus tie-policy head. Source: [results/paper_tables/antmaze_rawobs_transition_tie_policy_head_ep10_tseed012.csv](#).

Environment	Controller	Evaluation	Bellman matched	Stochastic TRL	Full Bellman	Transition top-1	Action agreement
AntMaze navigate	50k full-goal BC + body-nearest k16	seed 0, tasks 4-5, 10 episodes/task	0.300	0.950	0.950	0.956	0.999
AntMaze stitch	20k full-goal BC + body-nearest k16	seed 0, tasks 4-5, 10 episodes/task	0.300	1.000	1.000	1.000	0.999
AntMaze navigate	50k full-goal BC + body-nearest k16	3 eval seeds, tasks 4-5, 10 episodes/task	0.283	0.933	0.933	1.000	0.999
AntMaze stitch	20k full-goal BC + body-nearest k16	3 eval seeds, tasks 4-5, 10 episodes/task	0.283	0.967	0.967	1.000	0.999

Table 17: AntMaze raw-observation transition plus previous-action policy head. Source: [results/paper_tables/antmaze_rawobs_transition_prev_policy_head_ep10_evalseed012.csv](#).

Environment	Controller	Evaluation	Bellman matched	Stochastic TRL	Full Bellman	Transition top-1	Action agreement
AntMaze navigate	50k full-goal BC + body-nearest k16	3 eval seeds, tasks 4-5, 10 episodes/task	0.350	0.933	0.933	1.000	1.000
AntMaze stitch	20k full-goal BC + body-nearest k16	3 eval seeds, tasks 4-5, 10 episodes/task	0.283	0.967	0.950	0.994	1.000

Table 18: PointMaze learned-controller screen.

Environment	Controller	Evaluation	Bellman matched	Stochastic TRL	Full Bellman
PointMaze navigate	5k full-goal BC + body-nearest k16	3 eval seeds, 20 episodes/task	0.323	1.000	1.000
PointMaze stitch	5k full-goal BC + body-nearest k16	3 eval seeds, 20 episodes/task	0.223	1.000	1.000

Table 19: Controller execution isolation.

Execution path	Environment	Actor/executor	Evaluation	Best	Final
GCFBC direct final-goal actor	PointMaze navigate	GCFBC 128,128	seed 0, 5 episodes/task	0.080	0.000
MSEBC direct final-goal actor	PointMaze navigate	MSEBC LN 256,256	seed 0, 5 episodes/task	0.120	0.120
Stochastic TRL + learned waypoint BC	PointMaze navigate	5k full-goal BC + body-nearest k16	3 eval seeds, 20 episodes/task	1.000	1.000
Stochastic TRL + learned waypoint BC	PointMaze stitch	5k full-goal BC + body-nearest k16	3 eval seeds, 20 episodes/task	1.000	1.000